ASSIGNMENT 2

SUBJECT CODE: CSU1658
SUBJECT NAME: Statistical Foundation of Data Sciences
TOTAL POINTS: 100
DEADLINE: NOVEMBER 14TH 2025 4PM

PROBLEM 1.

Let X be a zero-inflated normal:

$$\mathbb{P}(X = 0) = \alpha, \quad X \mid (X \neq 0) \sim \mathcal{N}(\mu, \sigma^2)$$

with prob. $(1 - \alpha)$.

1. Write the PMF/PDF and CDF of X.
2. Derive $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.
3. Numerically evaluate $\mathbb{P}(X \leq 0)$ for $\alpha = 0.3$, $\mu = 2$, $\sigma = 1$.

PROBLEM 2.

Let $X \sim \mathrm{Bin}(n, p)$ with $n = 10$, $p = 0.3$.

1. Compute exactly $\mathbb{P}(X \geq 6)$.
2. Approximate $\mathbb{P}(X \geq 6)$ by a normal approximation with continuity correction and compare.

PROBLEM 3.

Suppose $N \sim \mathrm{Poisson}(\lambda)$. Conditional on N, each of the N events is independently tagged "A" with probability q (and "B" otherwise). Let K be the number of "A" events.

1. Show $K \sim \mathrm{Poisson}(\lambda q)$ (the thinning property).
2. Show that, conditional on $N = n$, $K \mid N = n \sim \mathrm{Bin}(n, q)$.
3. For $\lambda = 12$, $q = 0.2$, compute $\mathbb{P}(K = 0)$ and $\mathbb{P}(K \leq 2)$ exactly (closed form).

PROBLEM 4.

Let $X = (X_1, \dots, X_n)$ be nonconstant real data with population mean $\mu = \frac{1}{n}\sum_i X_i$ and population standard deviation $\sigma = \sqrt{\frac{1}{n}\sum_i (X_i - \mu)^2}$. Define:

- **z-score standardization:** $Z_i = \frac{X_i - \mu}{\sigma}$.
- **min-max normalization to** $[0, 1]$**:** $M_i = \frac{X_i - \min X}{\max X - \min X}$.
- **robust scaling (median/IQR):** $R_i = \frac{X_i - \mathrm{med}(X)}{\mathrm{IQR}(X)}$.

(a) Prove for z-scores that $\bar{Z} = 0$ and $\mathrm{Var}(Z) = 1$.

(b) Show that Pearson correlation is invariant to positive affine transforms of either variable: for constants $a > 0, b \in \mathbb{R}$.

$$\rho(X, aY + b) = \rho(X, Y), \qquad \rho(aX + b, Y) = \rho(X, Y).$$

(For $a < 0$, ρ flips sign.)

(c) Give the inverse transforms to recover X from Z and M.

(d) Numerical check on $x = (3, 7, 7, 19, 21)$.

PROBLEM 5.

You collect $n = 200$ observations: 199 are approximately $N(0, 1)$ and one extreme outlier at value 8. Compare three binning rules:

- **Sturges:** $K = \lceil \log_2 n \rceil + 1$ bins (equal width).
- **Scott:** bin width $h_S = 3.5\, s\, n^{-1/3}$, where s is the sample standard deviation.
- **Freedman-Diaconis (FD):** $h_{FD} = 2\, \mathrm{IQR}\, n^{-1/3}$.

Assume the 199 near-normal points have mean 0 and variance 1 exactly (so $\sum_{i=1}^{199} x_i = 0$ and $\sum x_i^2 = 199$) and add $x_{200} = 8$.

(a) Compute the new mean and population variance s^2.

(b) Compute h_S and h_{FD}. (Use IQR ≈ 1.349 for a standard normal: a single outlier scarcely affects IQR.)

(c) Suppose you display data on $[-4, 8]$ (range $= 12$). Report the **effective bin counts** under Sturges, Scott, and FD, and interpret robustness.

PROBLEM 6.

Two groups, $X \in \{0,1\}$, with outcomes $Y \in \{0,1\}$. Counts:

	$Y=1$	$Y=0$
$X=0$	a	b
$X=1$	c	d

Model: $\text{logit}\{P(Y=1|X)\} = \beta_0 + \beta_1 X$.

(a) Show

$$\hat{\beta}_0 = \log\frac{a}{b}, \quad \hat{\beta}_1 = \log\frac{c/b}{a/d} = \log\frac{cb}{ad}$$

(provided all cells ¿ 0).

(b) Show $\text{Var}(\hat{\beta}_1) \approx \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$ and give a 95% Wald CI for $\exp(\beta_1)$ (odds ratio).

(c) Numerical example: $a = 18, b = 12, c = 30, d = 10$.

PROBLEM 7.

Five subjects each measured under three conditions (within-subject). Subject means and condition means are:

Raw data (responses):

Subject	$C1$	$C2$	$C3$
1	10	12	13
2	9	11	12
3	8	10	11
4	9	10	12
5	11	12	14

Let's test the condition effect with the standard within-subjects ANOVA (assuming sphericity).

PROBLEM 8.

For binary class probability $p \in [0,1]$, define

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p), \quad G(p) = 2p(1-p), \quad E(p) = \min\{p, 1-p\}.$$

(a) Prove $H(p) \geq G(p)$ for all p.

(b) Prove $G(p) \geq E(p)$ for all p.

(c) Provide a small table of values to compare scales.

PROBLEM 9.

You have 12 training rows with binary target $Y \in \{0, 1\}$ (6 positives, 6 negatives). Consider two candidate attributes:

- **Attribute A** has three values v_1, v_2, v_3 with class counts:

A level	Count	Positives	Negatives
v_1	5	4	1
v_2	3	1	2
v_3	4	1	3

- **Attribute B** is binary with class counts:

B level	Count	Positives	Negatives
b_0	6	4	2
b_1	6	2	4

1. Compute the **parent entropy** $H(S)$ and **parent Gini** $G(S)$.
2. For A, compute the **weighted entropy after split**, the **Information Gain** $\mathrm{IG}(S; A)$, the **weighted Gini after split**, and the **Gini decrease**.
3. Repeat (2) for B.
4. Which attribute would ID3 (entropy) and CART (Gini) choose?

PROBLEM 10.

Data ($n = 5$):

x	0	1	2	3	4
y	1.0	2.9	5.1	5.9	9.2

Fit the simple linear model $y = \beta_0 + \beta_1 x + \epsilon$.
Tasks

1. Compute $\hat{\beta}_1$, $\hat{\beta}_0$.
2. Compute residuals, SSE, R^2, residual standard error s.
3. Test $H_0 : \beta_1 = 0$ (two-sided) and give a 95% CI for β_1.
4. Give a 95% prediction interval for y at $x_0 = 5$.